

F.7.9 Zanobetti and Schwartz (2008b)

Zanobetti and Schwartz (2008b) evaluated the relationship between short-term ozone exposure and all-cause mortality across 48 U.S. cities (1989-2000) for all ages in a case-crossover study. Data on 6,951,395 total deaths was provided by the National Center for Health Statistics. Daily 8-hour mean O₃ concentrations were obtained from the U.S. EPA Air Quality System Technology Transfer Network for all seasons. The authors ran conditional logistic regression models by season, month, and age group (0-20, 21-30, 31-40, 41-50, 51-60, 61-70, 71-80, 80+) controlling for temperature, dew point, and day of the week.

Mortality, All Cause

In single-pollutant models, the coefficient and standard error are estimated from an excess risk of 0.08% (-0.42-0.57%) for ages 0-20; 0.1% (-0.67-0.87%) for ages 21-30; 0.07% (-0.38-0.52%) for ages 31-40; 0.08% (-0.27-0.43%) for ages 41-50; 0.54% (0.19-0.89%) for ages 51-60; 0.38% (0.16-0.61%) for ages 61-70; 0.5% (0.32-0.67%) for ages 71-80; and 0.29% (0.13-0.44%) for ages 81-99 for a 10 ppb increase in average daily 8-hour mean all season ozone (Zanobetti and Schwartz 2008b, Table 2).

Appendix G. Additional Health Impact Functions in U.S. Setup

In this Appendix, we present additional health impact functions for estimating PM_{2.5} and Ozone-related adverse health effects. Unlike Appendices E and F, these functions are included in the U.S. Setup but are not currently used by the U.S. EPA in regulatory impact analyses. For the health impact functions currently used by EPA, see the following page: <https://www.epa.gov/benmap/benmap-community-edition>. For Ozone Health Impact Functions, click the “U.S. EPA approach for quantifying and valuing ozone effects” link. For PM_{2.5} Health Impact Functions, click the “U.S. EPA approach for quantifying and valuing PM effects” link.

G.1 Additional PM_{2.5} Health Impact Functions

Section G.1 summarizes the additional health impact functions for PM_{2.5} included in BenMAP-CE.

G.1.1 Long-term Mortality

There are two types of exposure to PM that may result in premature mortality. Short-term exposure may result in excess mortality on the same day or within a few days of exposure. Long-term exposure over, say, a year or more, may result in mortality in excess of what it would be if PM levels were generally lower, although the excess mortality that occurs will not necessarily be associated with any particular episode of elevated air pollution levels. In other words, long-term exposure may capture a facet of the association between PM and mortality that is not captured by short-term exposure. Table G-1 lists the additional long-term mortality health impact functions.

Table G-1. Additional Health Impact Functions for Particulate Matter and Long-Term Mortality*

Effect	Author	Year	Location	Age	Metric	Beta	Std Err	Form	Notes
Mortality, All Cause	Expert A	2006		30-99	Annual	0.015180		Log-linear	
Mortality, All Cause	Expert B	2006		30-99	Annual	0.012620		Log-linear	Range >10 to 30 µg. Unconditional dist. 2% no causality included.
Mortality, All Cause	Expert B	2006		30-99	Annual	0.011950		Log-linear	Range 4 to 10 µg. Unconditional dist. 2% no causality included.
Mortality, All Cause	Expert C	2006		30-99	Annual	0.011930		Log-linear	
Mortality, All Cause	Expert D	2006		30-99	Annual	0.008380		Log-linear	Unconditional dist. 5% no causality included
Mortality, All Cause	Expert E	2006		30-99	Annual	0.019670		Log-linear	Unconditional dist. 1% no causality included

Effect	Author	Year	Location	Age	Metric	Beta	Std Err	Form	Notes
Mortality, All Cause	Expert F	2006		30-99	Annual	0.011440		Log-linear	Range >7 to 30 µg
Mortality, All Cause	Expert F	2006		30-99	Annual	0.009370		Log-linear	Range 4 to 7 µg
Mortality, All Cause	Expert G	2006		30-99	Annual	0.006970		Log-linear	Unconditional dist. 30% no causality included
Mortality, All Cause	Expert H	2006		30-99	Annual	0.008700		Log-linear	
Mortality, All Cause	Expert I	2006		30-99	Annual	0.011810		Log-linear	Unconditional dist. 5% no causality included
Mortality, All Cause	Expert J	2006		30-99	Annual	0.009620		Log-linear	
Mortality, All Cause	Expert K	2006		30-99	Annual	0.006890		Log-linear	Range >16 to 30. No threshold. Conditional dist.
Mortality, All Cause	Expert K	2006		30-99	Annual	0.003940		Log-linear	Range 4 to 16 µg. No threshold. Conditional dist.
Mortality, All Cause	Expert K	2006		30-99	Annual	0.003940		Log-linear	Range 4 to 16 µg. Threshold 0 to 5 µg. Conditional dist.
Mortality, All Cause	Expert K	2006		30-99	Annual	0.003940		Log-linear	Range 4 to 16 µg. Threshold 5 to 10 µg. Conditional dist.
Mortality, All Cause	Expert K	2006		30-99	Annual	0			Range >16 to 30. Uniform distribution. No causality.
Mortality, All Cause	Expert K	2006		30-99	Annual	0			Range 4 to 16 µg. Uniform distribution. No causality.
Mortality, All Cause	Expert L	2006		30-99	Annual	0.009340		Log-linear	Range >10 to 30 µg. Unconditional dist. 1% no causality included.
Mortality, All Cause	Expert L	2006		30-99	Annual	0.007390		Log-linear	Range 4 to 10 µg. Unconditional dist. 25% no causality included
Short-term Mortality, Non-Accidental	Baxter et al.	2017	77 U.S. cities	0-99	D24HourMean	0.000329	0.000102	Log-linear	PM PA
Short-term Mortality, All Cause	Ito et al.	2013	150 U.S. cities	0-99	D24HourMean	0.000145	0.000075	Log-linear	PM PA
Mortality, IHD	Jerrett et al.	2017	Nationwide	30-99	Annual	0.013976	0.001775	Log-linear	PM PA

Effect	Author	Year	Location	Age	Metric	Beta	Std Err	Form	Notes
Mortality, All Cause	Krewski et al.	2009	116 U.S. cities	30-99	Annual	0.005827	0.000963	Log-linear	
Mortality, IHD	Krewski et al.	2009	116 U.S. cities	30-99	Annual	0.021511	0.002058	Log-linear	Mortality, IHD
Mortality, Lung Cancer	Krewski et al.	2009	116 U.S. cities	30-99	Annual	0.013103	0.003795	Log-linear	Mortality, Lung Cancer
Mortality, All Cause	Laden et al.	2006	6 cities	25-99	Annual	0.014842	0.004170	Log-linear	
Mortality, All Cause	Lepeule et al.	2012	6 Cities	25-99	Annual	0.013103	0.003347	Log-linear	
Mortality, All Cause	Pope et al.	2002	51 cities	30-99	Annual	0.005827	0.002157	Log-linear	
Mortality, IHD	Pope et al.	2015	Nationwide	30-99	Annual	0.013103	0.001791	Log-linear	PM PA
Mortality, All Cause	Thurston et al.	2016	100 U.S. metro areas	55-85	Annual	0.002956	0.001245	Log-linear	PM PA
Mortality, Lung Cancer	Turner et al.	2016	Nationwide	30-99	Annual	0.008618	0.003032	Log-linear	PM PA
Mortality, All Cause	Woodruff et al.	1997	86 cities	Infant	Annual	0.003922	0.001221	Logistic	
Mortality, All Cause	Woodruff et al.	2006	204 counties	Infant	Annual	0.006766	0.007339	Logistic	
Short-term Mortality, All Cause	Zanobetti et al.	2014	180 U.S. counties	65-99	D24HourMean	0.000638	0.000109	Log-linear	PM PA

*Unless specified as short-term mortality.

G.1.1.1 Expert Functions

In this section, we describe the approach taken to incorporate into BenMAP concentration- response (C-R) functions that were obtained through expert elicitation for EPA (IEc, 2006).

We have specified expert distributions for the PM_{2.5} effect either as truncated parametric distributions or as non-parametric distributions. Therefore they can only be included in BenMAP in the form of custom distribution tables containing 15,000 random draws (with replacement) from an underlying distribution. We first describe the way these custom distribution tables were created. Then we explain how these custom distribution tables should be handled in a configuration file to represent the expert-specified distribution as closely as possible.

Note that the table on page 3-30 of the expert elicitation report (IEc, 2006) refers to the non-parametric distributions as “custom” distributions. However, BenMAP refers to distribution tables that are supplied in the form of a simulated draw as “custom distribution tables”. In order to avoid confusion in terminology, we will call the expert-